

BIOMASS GASIFICATION

(LECTURE-9)



Views of Typical Biomass Gasification (Power) Plant (Source:BioEnergy Consult, July 2019)

BIOMASS GASIFICATION

LECTURE-9

➤ INTRODUCTION

- It is a process converting solid biomass fuels into a combustible gaseous fuel (or gas) called producer gas through a sequence of thermochemical reactions
- It produces a low heating value fuel with calorific values ranging from 1000 – 1200 kcal/m³
- Gas contains various percentage of CO, H₂, CO₂, CH₄ and N₂ depending upon the quality of feedstock (biomass) used, thermochemical conditions and the type of technology applied
- Overall reactions during biomass gasification may be shown by the following **Figure-1**:

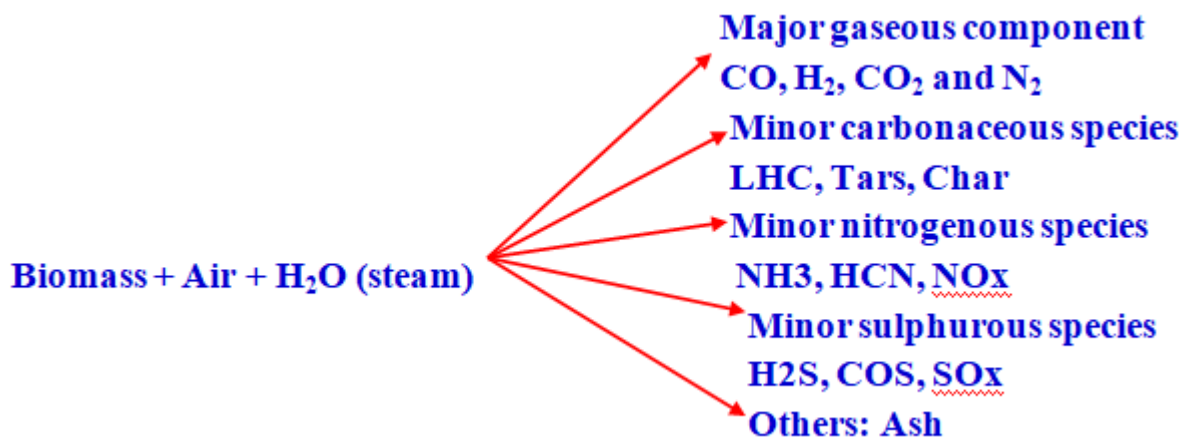


Figure-1

- Biomass gasification is one of the attractive technologies producing hydrogen rich gas, i.e. syngas
- Biomass gasification offers viable solution to overcome energy shortage
- Feedstocks predominated by carbon, hydrogen, oxygen and nitrogen are oxidised at elevated temperature to generate heat energy
- Heat produced during oxidation drives further chemical reactions to generate carbon monoxide, carbon dioxide, hydrogen and methane
- Different types of gasifiers are used depending on end use and quality of producer gas needed

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➤ Biomass Feedstock for Gasification

- biomass feedstocks are classified based on several factors, such as : moisture content, material, and form as shown in **Table-1**.
- biomass is broadly divided into three groups: dry, wet, and others
- dry biomass is classified as woody or herbaceous and wet biomass as sludge/excreta, common food, or other
- each classification has three sub-categories: waste, unutilized, and produced

Table-1: Classification of Biomass Feedstocks

Moisture	Classification	Wastes	Unutilised	Produced
Dry	Woody	Construction wastes, Timber offcuts	Forest thinnings, Remaining forest timbers, Damaged trees	Short-rotation woody crops (eucalyptus, willow, etc.)
	Herbaceous		Crop residues (rice / wheat straw, rice husk)	Grasses (Napier grass, sorghum, Miscanthus etc.)
Wet	Sludge /Excreta	Sewage sludge, Livestock excreta		
	Common food	Food-processing wastes, Kitchen wastes		
Other	Other	Molasses, Waste food oil	Landfill gas	Cultivated maize, Cultivated sugar cane

- among these, dry woody/herbaceous biomasses are used as feedstocks for BG plants in Japan
- woody biomasses contain waste woods (construction wastes and timber off cuts) and unutilized woods (forest thinning, remaining timbers, and damaged trees) composed of cedar, cypress, pine, etc.
- short-rotation woody crops (eucalyptus, willow, etc.) are categorized as produced woods
- energy crops such as willow may be cultivated in land fallow and used as biomass
- for unutilized herbaceous biomass, crop residues such as rice/wheat straw and rice husks
- while for produced herbaceous biomass, grasses such as Napier grass, sorghum, and Miscanthus are usable
- above all, rice husks contain abundant silica
- the ash byproduct from gasifiers has potential use in nanomaterials

➤ Chemistry of Gasification

- the reactions taking place in the gasifier can be summarized as given below:

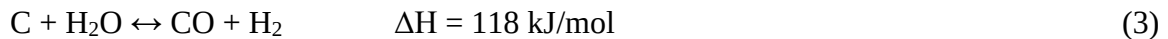
Partial oxidation:



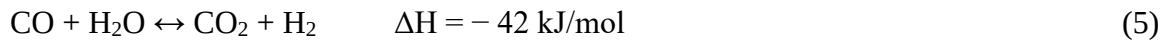
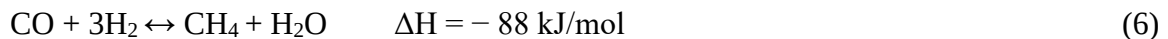
Complete oxidation:



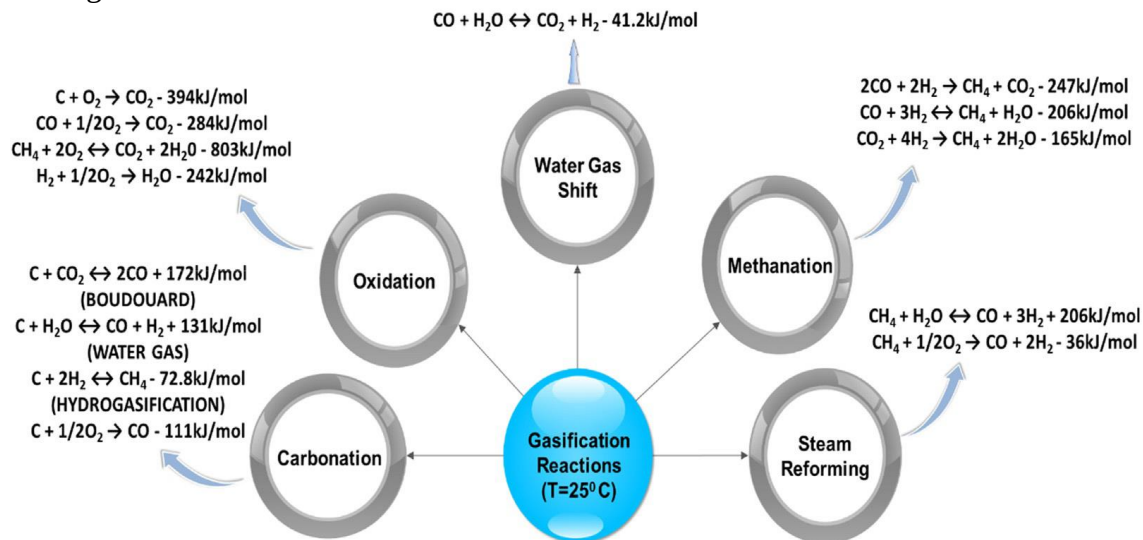
Water gas phase reaction:

**Boudouard reaction:**

- the heat required for water gas phase and Boudouard reactions is provided by complete and partial oxidation reactions, and complete oxidation provides around 60% of the heat requirements during gasification
- in addition to the previous reactions that are common in combustion and gasification, hydrogen, steam, and carbon monoxide undergo further reactions as shown below:

Water gas shift reaction:**Methane formation:**

- the water gas shift and methane formation reactions are in equilibrium and the governing parameters are: pressure, temperature, and concentration of reaction species
- the possible gasification reaction may also be summarized and shown by the following diagram

**➤ Overall Steps Involved in Biomass Gasification**

- the main steps involved can be categorized as upstream processing, the gasification process in the gasifier and the downstream processing as shown in **Figure-2**

Upstream Processing

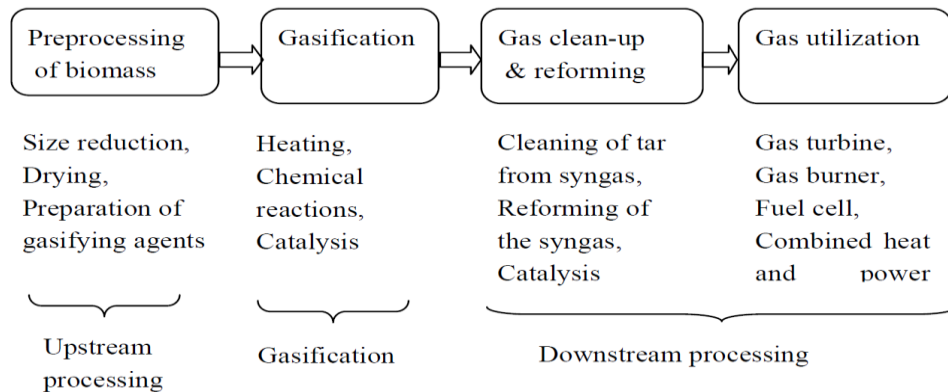
- upstream processing includes processing of the biomass to make it suitable for gasification operations
- size reduction is needed to obtain appropriate particle sizes
- drying is needed to achieve appropriate moisture so that the process can work efficiently
- densification also may be necessary due to the low density of biomass

Gasification

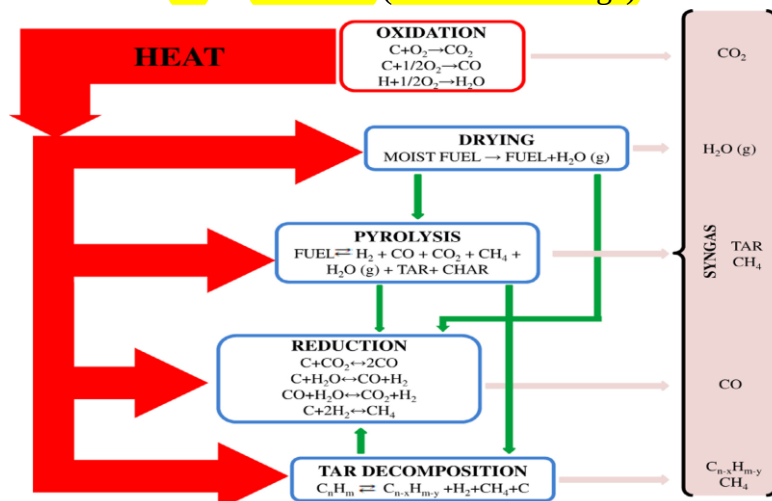
- this is the second step where the gasification process is carried out under specified conditions in order to achieve optimum product yield, i.e. **the producer gas**
- different types of gasifiers are used for the purpose

Down Stream Processing

- this is the third step where the cleaning (purification) of the gaseous products take place as per the desired specifications
- unit operations are applied for the purpose

**Figure-2: Processes involved in overall biomass gasification.****➤ The Biomass Gasification Process (Mechanisms of Gasification)**

- the biomass gasification process consists in the conversion of a solid/liquid organic compound in a gas/vapor phase and a solid phase
- the principal reactions of gasification are endothermic and the necessary energy for their occurrence is, generally, granted by the oxidation of part of the biomass, through an allo-thermal or an autothermal phase
- in the auto-thermal process, the gasifier is internally heated through partial combustion, while in the allo-thermal process the energy required for the gasification is supplied externally
- considering the auto-thermal system, gasification can be seen as a sequence of several stages
- a simplified schematic representation of the gasification is demonstrated in **Figure-3**
- the main steps of the gasification process are:
 - oxidation (exothermic stage)
 - drying (endothermic stage)
 - pyrolysis (endothermic stage)
 - reduction (endothermic stage)

**Figure-3: Main stages of the gasification process.**

- **Oxidation**

- ✓ in order to obtain the thermal energy required for the endothermic reaction and to maintain the operative temperature, the oxidation of part of the biomass is necessary
- ✓ the oxidation is carried out in conditions of **lack of oxygen with respect to the stoichiometric requirement**
- ✓ despite the partial oxidation involving all carbonaceous species (tars included), it is possible to simplify the system considering that only char and the hydrogen contained in the gas produced participate in the partial oxidation
- ✓ the main reactions that take place during the oxidation phase are given in **Figure-2** and also discussed in the chemistry of biomass gasification
- ✓ as evident, the main product of this step is the thermal energy necessary for the whole process, while the combustion product is a gas mixture of CO, CO₂ and water
- ✓ in this mixture nitrogen can be present if the biomass oxidation is performed with air, otherwise nitrogen is practically absent if only oxygen is used

- **Drying**

- ✓ drying consists in the **evaporation of the moisture** contained in the feedstock
- ✓ the amount of heat required in this stage is proportional to the feedstock moisture content
- ✓ generally, the heat required derives from the other stages of the process
- ✓ drying can be considered complete when a biomass temperature of **150 °C is achieved**

- **Pyrolysis**

- ✓ pyrolysis leads to the thermochemical decomposition of the matrix carbonaceous materials
- ✓ the cracking of chemical bonds takes place with the formation of molecules with a lower molecular weight
- ✓ by pyrolysis, it is possible to obtain different fractions: a solid, a liquid/condensed gas and non-condensed gas
- ✓ the solid fraction, which can range from 5–10 wt% for fluidized bed gasifiers to 20–25 wt% for fixed bed gasifiers, has a high carbon content and is characterized by a high heating value
- ✓ the solid fraction includes the inert materials contained in the biomass in the form of ashes and a high carbon content fraction, called “char”
- ✓ the liquid fraction, usually called "tars", varies according to the gasifier type, such as lower than 1 wt% for downdraft gasifiers, 1– 5 wt% for bubbling bed gasifiers, 10–20 wt% for updraft gasifiers and is constituted by complex organic substances, condensable at relatively low temperatures
- ✓ **the gaseous fraction is typically 70–90 wt% of the fed material and is a mixture of gases that are non-condensable at ambient temperature**
- ✓ the gaseous fraction is called "**pyrolysis gas**" and consists mainly of **hydrogen, carbon monoxide, carbon dioxide and light hydrocarbons** such as **methane and other C₂, C₃ hydrocarbons**; minor constituents are acid or inert gases
- ✓ the pyrolysis reactions take place with a temperature in the range **250–700 °C**
- ✓ they are endothermic and, as in the drying step, the heat required comes from the oxidation stage of the process
- ✓

- **Reduction**
 - ✓ the reduction step involves all the products of the preceding stages of pyrolysis and oxidation
 - ✓ the gas mixture and the char react with each other resulting in the formation of the final gaseous mixture
 - ✓ main reactions occurring in the reduction step have been shown in 'Chemistry of Biomass Gasification' and also demonstrated in **Figure-4**
- **Conceptual diagram** with respect to the mechanism of gasification demonstrated in multiple steps fixed-bed (a) updraft and (b) downdraft gasifiers

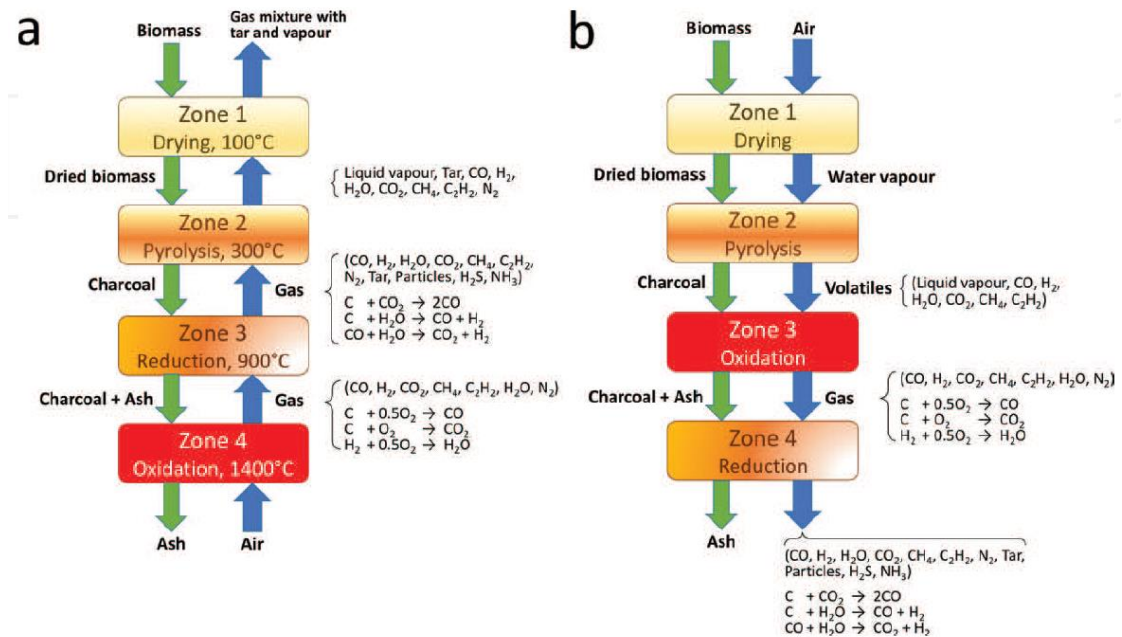


Figure-4: Conceptual diagram of multiple steps in fixed-bed (a) updraft and (b) downdraft gasifiers

➤ Flowchart for Biomass Gasification

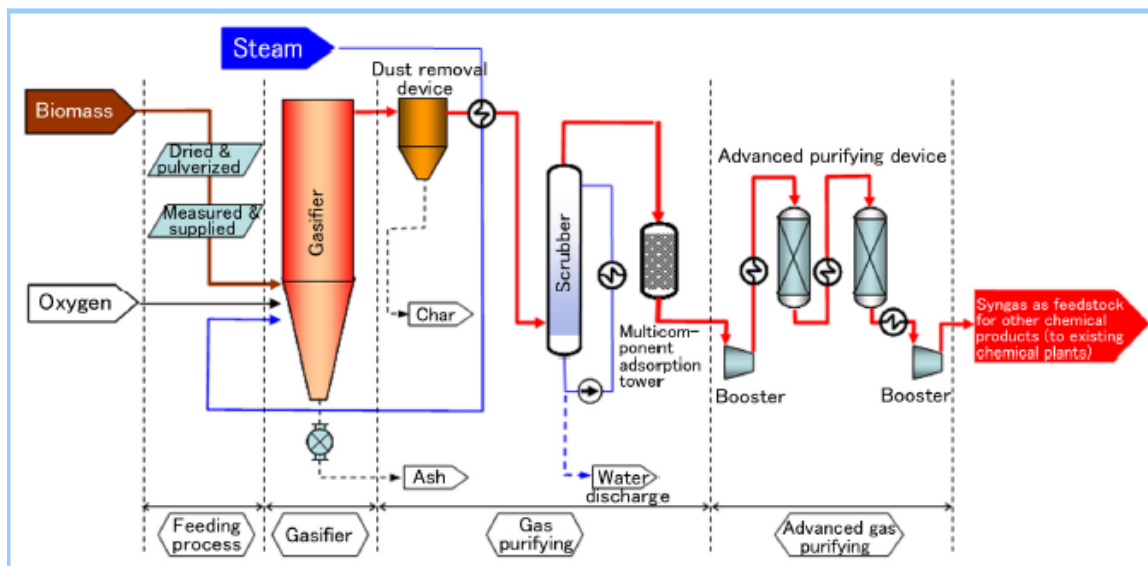


Figure 4 Diagram of the biomass gasification and syngas purifying system.

➤ Factors Affecting the Gasification Process

- the biomass gasification units may be divided into small-, medium-, and large-scale biomass gasification and power generation units
- pretreatment of biomass is needed which includes size reduction, size screening, separation of magnetic materials, and storing as wet biomass
- then prior to gasification, drying and storing as dry material are accomplished to reduce the moisture content
- feedstock type and feedstock preparation are important factors affecting the yield and quality of produced gaseous mixture or the syngas
- shredding and drying are two processes conducted to prepare the biomass raw material for gasification process
- the main parameters affecting the gasification are: equivalence ratio (ER), biomass characteristics, moisture content, moisture content, Superficial velocity, Operating temperature, gasifying agent, residence time, pressure, catalyst, effect of biomass/steam ratio

• Equivalence Ratio (ER)

- ✓ the equivalence ratio (ER), Φ is defined as the ratio of the actual air supplied to that of the stoichiometric/theoretical air used for gasification, i.e.

$$\Phi = \frac{\text{Actual Air supplied to the System for Combustion}}{\text{Theoretical/Stoichiometric requirement of Air for Combustion}}$$

- ✓ the efficiency, η of a gasifier is defined as

$$\eta = \frac{\text{Chemical energy output in the dry Producer Gas at } 15^\circ\text{C}}{\text{Energy input from the Biomass}}$$

- ✓ it may be seen from the following Figure-5 that total energy in the gaseous phase increases with the increase of equivalence ratio, Φ .

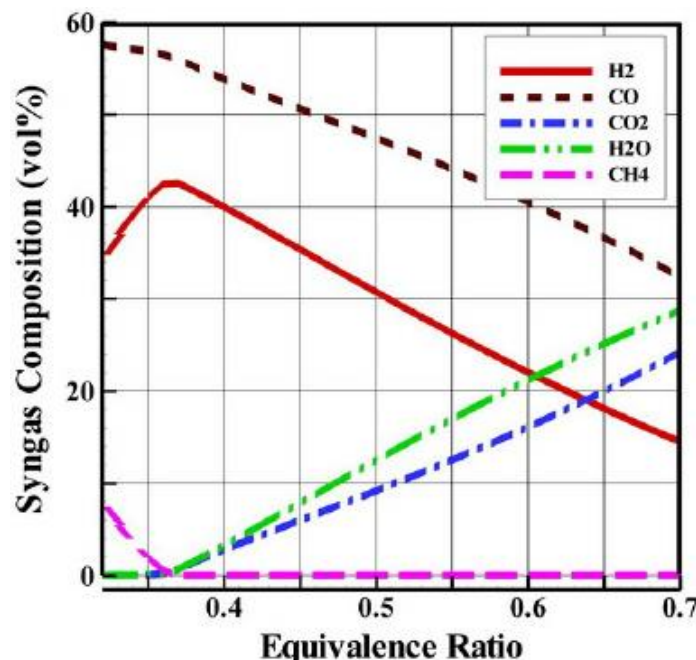


Figure-5: Producer gas composition with variable stoichiometric variation.

- ✓ increasing ER decreases the heating value of the produced gas due to decreasing H_2 and CO concentration and increasing CO_2 concentration
- ✓ higher ER helps in reducing tars and provides more O_2 to react with volatiles
- ✓ typical values of ER ranges between 0.2 and 0.4.
- ✓ it was reported that increasing ER decreases the concentration of combustible gases CO, CH_4 , and C_nH_m , however H_2 increases till the value goes to 0.4 and then it decreases
- ✓ increasing ER improves the reaction temperature and carbon conversion, and reduces the tar yield
- ✓
- **Biomass Characteristic**
 - ✓ biomass characteristic is a major factor affecting produced gases during gasification
 - ✓ the physical properties that may have major effect are: absolute and bulk density, and particulate size
 - ✓ the chemical composition parameters that are of major importance to define the quality including volatile matter, moisture content, fixed carbon, ash content, and gross calorific
 - ✓ the ultimate analysis comprises the carbon, oxygen, nitrogen, and sulfur of the dry biomass on a weight% are also highly significant
- **Moisture Content**
 - ✓ the moisture content can be determined by complete drying of biomass sample
 - ✓ the moisture content is calculated by subtracting the sample weight after drying from fresh sample weight
 - ✓ maximum allowable moisture content in downdraft gasifier is 40% on dry weight basis
 - ✓ updraft gasifier can handle biomass with higher moisture content
 - ✓ the higher moisture content in biomass will increase the consumed energy for drying, and will reduce the pyrolysis of biomass
 - ✓ as a general rule, increasing moisture content decreases the conversion
- **Superficial Velocity**
 - ✓ it is the ratio of the biogas production rate at normal conditions and the narrowest cross-sectional area of gasifier
 - ✓ lower superficial velocity is linked with high yield of char, and large quantities of unburned tars, which may deactivate catalyst, plug lines, and destroy compressors
 - ✓ higher superficial velocity results in reduced amount of char and low overall process efficiency
- **Operating Temperature**
 - ✓ operating temperature affects conversion, tar content, gas composition, gas heating value, and char conversion
 - ✓ in order to select the optimum temperature, gasifier type, and biomass source is considered
 - ✓ generally, temperature higher than $800^\circ C$ should be used to obtain high conversion and low tar content in the produced biogas
 - ✓ low temperature is associated with low tar content, low H_2 and CO content in the produced biogas
 - ✓ increasing temperature will increase gas yield, hydrogen, heating value, and ash agglomeration

- ✓ the ash agglomeration problem can be overcome by keeping the temperature below 750°C
- **Gasifying Agent**
 - ✓ gasifying agents in use are air, steam, steam/oxygen mixture, and CO₂
 - ✓ they affect the heating value of the produced gas
 - ✓ the heating value increases with increasing steam content of the gasifying agent, whereas heating value decreases as air increases in the gasifying agent
 - ✓ the steam/oxygen mixture represents a zero nitrogen-gasifying agent which increases the heating value and allows liquefying the produced gas after proper treatment
- **Residence Time**
 - ✓ it has a significant impact on the composition and produced tars in biomass gasification
 - ✓ oxygen-containing compounds may be decreased by increasing the residence time
 - ✓ increasing residence time decreases yield of one and two atomic ring compounds,
- **Pressure**
 - ✓ in gasification processes, either atmospheric or and higher pressures are commonly used
 - ✓ selection of the optimum pressure depends on the application of the produced gaseous product
 - ✓ if it is used for production of methanol or synthetic auto-fuels, higher pressures are preferred to improve the process yield and to reduce tar content
 - ✓ for generating burnable gases, atmospheric pressure should be used
 - ✓ high pressure applications are recommended for large scale gasification
 - ✓ however, atmospheric pressure is recommended for small-scale gasification
 - ✓ high pressure gasification is still not well developed and further research is needed to commercialize such processes
- **Catalyst**
 - ✓ The type of the catalyst is an important factor affecting gasification quality and the product
 - ✓ catalyst affects the composition of the syngas by manipulating the percentage volume of hydrogen, carbon dioxide, methane, and carbon monoxide
 - ✓ optimum quantity of catalyst plays a significant role in minimizing the gas content of carbon dioxide and maximizing the useful gases in the gaseous products such as hydrogen, carbon monoxide, and methane

➤ **Technologies of Biomass Gasification**

- **Introduction**
 - ✓ gasification process converts biomass, a low-energy density material, into a gaseous product mixture containing mainly CO, H₂, CH₄ and CO₂.
 - ✓ gasification is a partial oxidation process and it is commonly operated at 800–900 °C.
 - ✓ in some cases, steam is also used as the gasification agents
 - ✓ the gaseous products from the gasifier can be utilized in gas engines or gas turbines for the generation of electricity
 - ✓ gasifier reactors are simple in construction, however the chemistry and physics involved in their operation is not well understood

- ✓ gasifiers designs are generally categorized into three types: downdraft, updraft and fluidized bed
- ✓ all of the gasification processes oxidize part of the biomass to generate energy and to carry out the process
- ✓ the selection of the gasifier depends on the end use and quantity of the producer gas required
- ✓ in fixed bed gasifier the biomass fuel is fixed or stable on the bed and does not move
- ✓ fixed-bed biomass gasifiers are classified into two category: down draft and updraft
- **Downdraft Gasifier**
 - ✓ in downdraft gasifier, fuel is fed near the top of the reactor
 - ✓ the schematic diagram of the system is shown in **Figure-6**
 - ✓ air can be introduced alongwith the fuel or at some intermediate level below
 - ✓ hot producer gas alongwith char and ash exit the bottom of the reactor vessel
 - ✓ a slight vacuum is usually applied at the gasifier exit to withdraw producer gas
 - ✓ the four different zones of drying, pyrolysis, combustion and reduction occur sequentially from the top to the bottom of the gasifier, respectively
 - ✓ the top section of the gasifier acts as a surge vessel for fuel to be dried before entering the pyrolysis zone
 - ✓ since the top of the downdraft gasifier is at ambient temperature and pressure, fuel feeding is quite simple

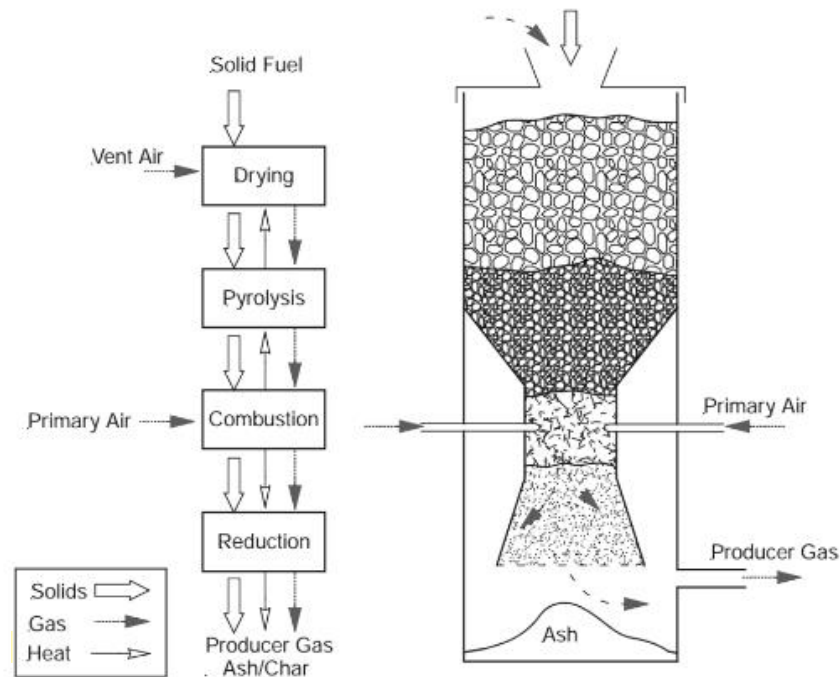


Figure-6: Downdraft gasifier.

- ✓ radiant and combusive heat transfer from the lower pyrolysis and the combustion sections provide the heat for drying
- ✓ water vapor from the drying zone flows downwards towards the pyrolysis and combustion section where a portion is reduced to hydrogen
- ✓ however, majority of water vapour remains in the final producer gas
- ✓ a small amount of air may be added in the drying section to facilitate the downward flow of gas, thereby sweeping out excess moisture

- ✓ the dried solid fuel moves downward by gravity into the pyrolysis zone and is heated such that volatile matter is evolved
- ✓ resulting gas flows through the reduction zone on an ash support grate before exiting the reactor
- ✓ gaseous products of the pyrolysis reactions are drawn downwards along with the char into the combustion zone of the gasifier
- ✓ a portion of the pyrolysis products and char are burned in the combustion zone as free oxygen becomes available
- ✓ at this point, an oxidizer gas, such as air or oxygen is injected
- ✓ reactions with oxygen results in steep rise of temperature upto $1000 - 1200^{\circ}\text{C}$, depending on the moisture content of the fuel
- ✓ some downdraft reactor designs employ a restriction, commonly known as a throat section, to increase the velocity of gas to promote heat and mass transfer
- **Updraft Gasifier**
 - ✓ updraft gasification system has been shown by **Figure-7**
 - ✓ air is injected near the bottom of the reactor (in the combustion zone) with the resulting producer gas flowing upwards through the interstitial spaces in the solid fuel
 - ✓ the producer gas exit the top of the reactor where incoming biomass fuel is added
 - ✓ hence the updraft producer gas exits at lower temperature ($130 - 150^{\circ}\text{C}$) than that of the downdraft reactor due to the drying of the solid fuel

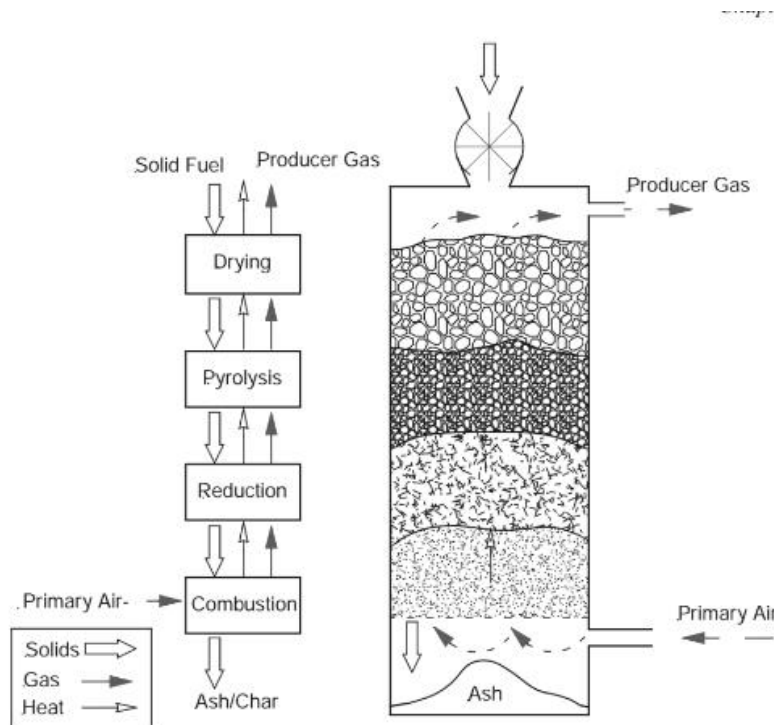


Figure-7: Updraft gasifier.

- ✓ however, the gas is heavily contaminated by pyrolysis products (primarily tar, oils and particulate) from the incoming fuel
- ✓ Since the producer gas exits at the top, special precautions are taken to avoid air in-leakage with the incoming biomass

- ✓ the updraft gasification process is similar to that of the downdraft mode in that the major reaction stages are present: drying, pyrolysis, combustion and reduction
- ✓ however, with respect to the solids, the order of these stages is changed
- ✓ most of the moisture evaporated from the fuel solids and products of the pyrolysis reactions exit the top of the reactor with the producer gas
- ✓ the charged solid migrates below to the reduction zone
- ✓ heated combustion gases enter the reduction zone providing the required energy for the endothermic reactions
- ✓ finally, the remaining char reacts with oxygen or air injected into the bottom of the reactor vessel
- ✓ ash is extracted from the bottom of the reactor
- **Fluidized-Bed Gasifier**
 - ✓ schematically, the fluidized-bed gasifier is shown in Figure-7
 - ✓ it makes the use of an inert medium such as sand to mix the solid fuel with the gas phase
 - ✓ depending on the specific reactor design and average bed media particle size, the superficial gas velocity typically ranges from 0.5 – 1.0 m/s
 - ✓ air is injected in the bottom of the reactor vessel through a distributor comprising of a perforated plate and series of nozzles such that the bed material is suspended in a fluid-like state as evident from the Figure –8, below:

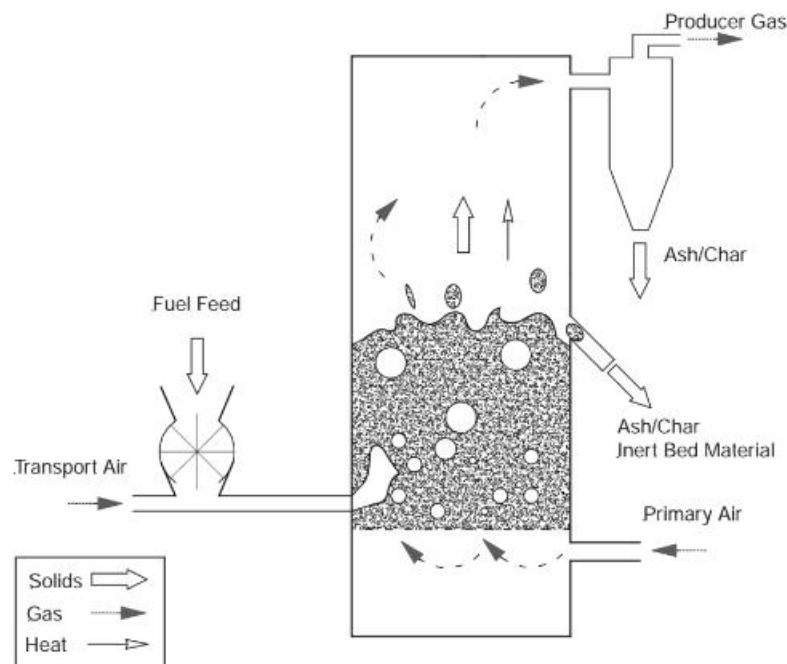


Figure-8: Fluidized-Bed Gasifier

- ✓ gas bubbles form at the distributor, increasing in size as they rise upward toward the bed surface due to the decreasing hydrostatic pressure
- ✓ air within the bubbles exchanges with combustion and pyrolysis gases formed in the dense phase as the fuel is converted to producer gas
- ✓ fly ash, char, and inert bed materials are ejected from the dense phase bed as gas bubbles erupt on the surface
- ✓ the turbulent fluidized medium promotes high heat-transfer rates, and as a result the bed-temperature is maintained in an essentially isothermal state

- ✓ the maximum temperature obtained during gasification is only 800 °C – 900 °C which is less than the temperature that of the fixed bed design (downdraft, updraft)
 - ✓ this is combined with the low gas residence time which increases the yields of tars, oils, and light hydrocarbons
 - ✓ the tar/oil concentration in the producer gas is higher than that derived from the downdraft gasifier
 - ✓ fuel is pneumatically conveyed into the bottom of the dense-phase bed
- **Entrained Flow Gasifier**
 - ✓ entrained flow gasifier is a comparatively new design with high efficiency
 - ✓ it is normally employed for large-scale gasification of coal, biomass, and refinery residues
 - ✓ however, fuel particles should be highly pulverized for this kind of gasifier, and hence it is problematic when biomass is used as a feedstock
 - ✓ on the other hand, gasification in these gasifiers takes place above 1000 °C which helps in cracking tar
 - ✓ they are basically classified into two families, namely, top-fed gasifier and side-fed gasifier as shown in the **Figure-9**, below:

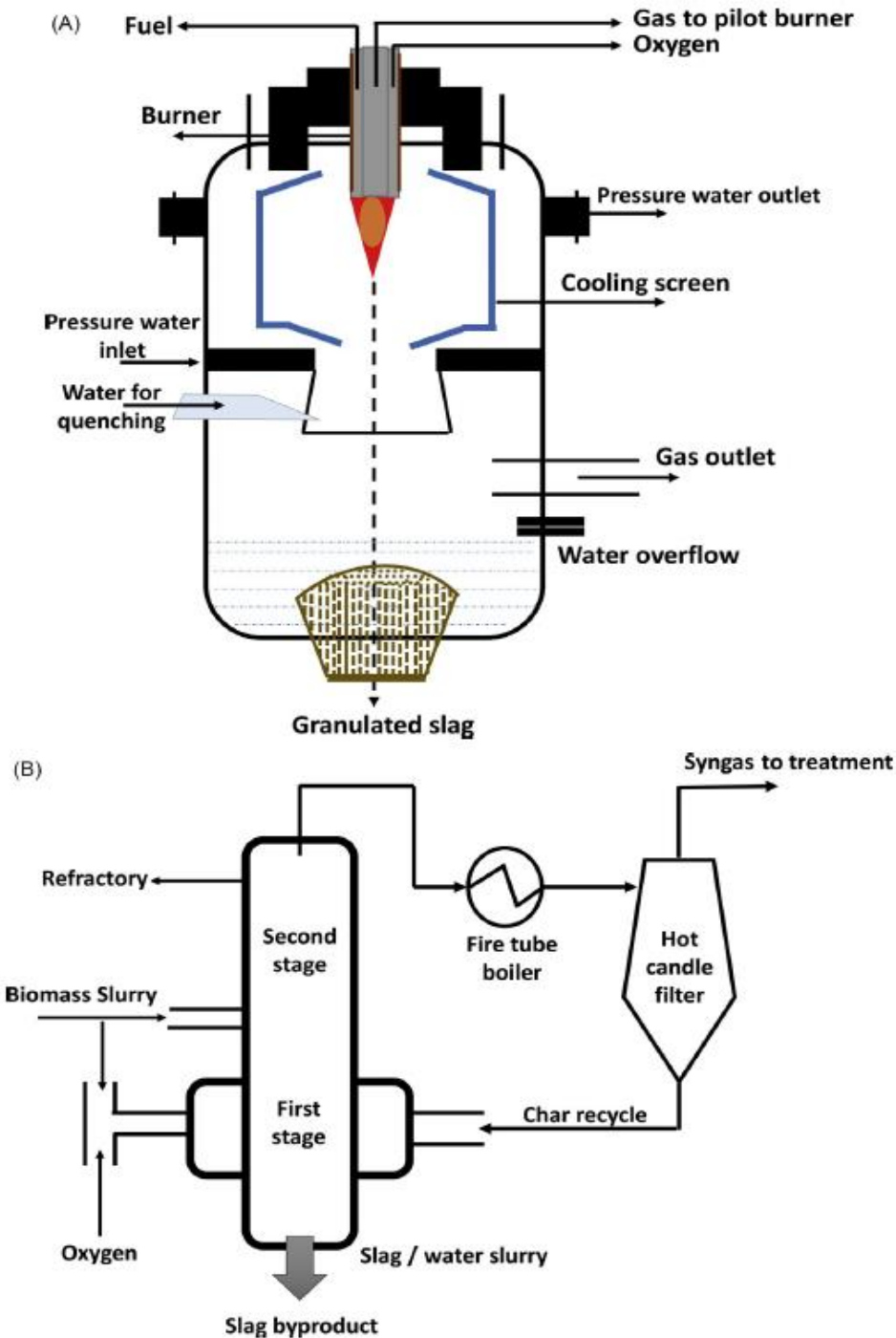


Figure-9: Top-fed entrained flow gasifier. (B) Side-fed entrained flow gasifier.

- ✓ top-fed entrained flow gasifiers are vertically aligned cylindrical shaped vessels
- ✓ finely refined fuel particles and gasifying media come in the form of a jet from the top end of the reactor (**Figure-8A**)
- ✓ an inverted burner results in their combustion followed by gasification
- ✓ product gas is taken out from the side of the lower section, whereas slag is deposited at the bottom of the reactor
- ✓ in side-fed gasifier, pulverized fuel and gasifying agent are fed through nozzles present in the lower part of the reactor (**Figure-9B**)
- ✓ this design results in appropriate mixing of fuel and oxygen

- ✓ the product gas is collected from the top and the slag from the bottom of the vessel
- ✓ other important issues that process designs need to deal with are slagging, fouling, and corrosion
- ✓ these issues arise out of the inorganic species present in the biomass and are, therefore, dependent to a large part on the biomass composition

➤ Advantages and Disadvantages of Various Biomass Gasification Technologies (Gasifiers) used

• Fixed Bed

- ✓ Illustrated in the Table-2, below:

Table-2: Advantages and Disadvantages of Fixed Bed Gasifiers

Reactor type	Advantage	Disadvantage
Fixed bed "Updraft"	-High thermal efficiency -Good contact between the solid material and the oxidizing agent -Can handle materials of different sizes	-High content of tar in the syngas -Energy content of tar > 20% -Low production of CO and H ₂ . -requires a subsequent treatment of the tar cracking -Limited flexibility to load and process (the treated material should have properties homogeneous)
	-Can handle materials with high humidity -Reduced entrainment both of dust and that of Ashes -Simple construction	-Reduced starting difficulties and temperature control -Need for installation of mobile grates to avoid the formation of preferential paths in the fixed bed -According to catalysts, they may be not usable since syngas energy may be lower than that necessary for the activation, requiring external energy supply -Poisoning deactivation of catalysts may be possible
Fixed bed "Downdraft"	-Robust technology -There are no problems of scale-up	-Low specific capacity -Need for uniform sizes in input (pellets no larger than 100 mm) -Training of sintered slag on the grid
	- High carbon conversion -Low production of tar -Limited entrainment of ash and dust -High solid residence time -Simple construction -Reliable technology	-Requires materials with a low moisture content -Limited flexibility to load and process (the treated material must have the same characteristics) -Low coefficient of heat transfer -Difficulty starting and controlling the temperature -Poisoning deactivation of catalysts may be possible -Limited possibility of scale-up (the dimensions of the reactor are limited by problems of temperature control)

• Entrained Flow Gasifier

- ✓ Reflected by the following Table-3:

Table-3: Advantages and Disadvantages of Entrained Flow Gasifiers

Reactor type	Advantage	Disadvantage
Entrained flow reactor	-Fuel flexibility -Uniform temperature -High carbon conversion -There are not problems of scale-up -Good ability to control the parameters process -Short reactor residence time. -Very low tar concentration. -High Temperature slagging operation (vitrified slag).	-Large oxidant requirements -High level of sensible heat in product gas -Heat recovery is required to improve efficiency -Low cold gas efficiency -Requires the reduction of size and preparation supply -Short life of system components, including gasifier vessel refractory -High plant cost -High maintenance cost

Table 6. Advantages and disadvantages of fixed bed reactors.

Reactor type	Advantage	Disadvantage
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- **Fluidized Bed Gasifier**

✓ Summarized in the Table-4, below:

Table-4: Advantages and Disadvantages of Fluidized Bed Gasifiers

Reactor type	Advantage	Disadvantage
Bubbling fluidized bed	-High mixing and gas-solid contact -High carbon conversion -High thermal loads -Good temperature control (temperature distribution along the reactor) -Can handle materials with different characteristics -Good flexibility both of load and process	-Loss of carbon in the ashes -Dragging of dust and ashes -Pre-treatment need with heterogeneous materials -Need to have a relatively low process temperature to avoid phenomena of de fluidization of the bed (temperature lower than the softening point of the solid residues)
	-Suitable for highly reactive fuels such as biomass and municipal waste pre-treated -Low level of tar in the syngas -Ease of start-up, shutdown and control -Possibility to use catalysts, even on a large scale (thanks to the good temperature control) -No moving parts -Good ability to scale-up	-Restrictions on the size -High investment costs and maintenance costs
Circulating fluidized bed	-Lower tar production -High conversions -Flexible load -Reduced residence times -Good ability to scale-up	-Possibility of casting the ashes -Loss of carbon in the ashes -Requires the reduction of size and preparation supply (the solid material must be finely pulverized, with dimensions lower than 100 mm) -Restricted solid-gas contact - Need for special materials -Technology complex and difficult to control. -Security issues -High start-up costs and investment costs